How the Governed Decision Engine Works

A proof-style walkthrough of the merchant financing engine

1 INPUTS

Merchant evidence and requested terms feed derived assessments.

- **Merchant context** — Industry and relationship; policy context.
- **Requested structure** — Amount, remittance, payback multiple and horizon.
- **Sales & cash flow** — Sales, settlement, deposits and usable history.
- **Obligations & liquidity** — Cash balances and existing payment commitments.
- **Verification & continuity** — Source confidence, verification and processor continuity.
- **Derived risk & economics** — Synthetic comparative risk, loss, costs and contribution.

Source inputs and derived assessments stay distinct; retain run and source identity.

2 ELIGIBILITY GATES

Different conditions require different routes—not one generic failure.

- **Evidence sufficiency** — Required inputs must be usable. Missing is not zero.
- **Policy hard stops** — Non-remediable conditions remain declines.
- **Structure conditions** — Capacity and economics can require a search.
- **Judgment requirements** — Defined review conditions can remain after search.
- **Advisories** — Retain warnings; apply the specific source rule.

Terms cannot repair missing evidence or waive hard stops.

3 STRUCTURE SEARCH

M2.2 constructs and tests. M2.3 resolves final authority.

- **Horizon** — 30 / 60 / 90 days
- **Remittance** — Sales-linked share
- **Amount** — Advance dollars
- **Payback multiple** — Repayment target factor
- **Economic hurdle** — Required modeled contribution test

POLICY GUARDRAILS

- Minimum counteroffer = 40% of requested amount
- Maximum amount reduction = 60%

Test admissible terms, credit and economics together; final authority remains separate.

4 DECISION CIRCUIT

Structure support and final authority are separate tests.

Structure support
Requested or alternative terms satisfy the applicable tests.

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Final authority / M2.3
Strict or review conditions + matched-scenario floor.

Support ≠ authorization.

ROUTES TO OUTCOMES

1. **Approve as requested** — Requested terms satisfy governing tests and authority.
2. **Counteroffer authorized** — Alternative terms satisfy governing tests and authority.
3. **Counteroffer review required** — Selected alternative; binding review requirements remain.
4. **Manual underwriting review** — Source-defined underwriting judgment remains required.
5. **Evidence decline** — Required inputs cannot support the dependent assessment.
6. **No feasible structure** — No eligible final structure after search and governance.
7. **Policy hard stop** — A non-remediable policy condition controls.

Retain the final outcome, applicable terms and binding reasons.

5 OUTPUT PROOF

Every decision retains reasons, source identity and supporting evidence.

- **Decision record & terms**
  - Final outcome and applicable governed terms.
  - A review path is not a funded offer.
- **Reason codes**
  - Standardized policy reasons
  - Consistent across teams
- **Explainable rationale**
  - Key drivers and trade-offs
  - Human-readable justification
- **Versioned data contract**
  - Typed outputs and consumption schema.
  - Retained source lineage.
- **Governed evidence trail**
  - Source and version identities.
  - Retained decision evidence.

Trace the outcome to its inputs, rules and retained evidence.

GOVERNED — Policy-driven rules and guardrails.

DETERMINISTIC — Same versioned inputs and policy → same outcome.

EXPLAINABLE — Clear reasons, transparent logic, human-readable.

REPRODUCIBLE — Versioned sources and retained evidence.

From first advance to intelligent portfolio decisions.

Deterministic synthetic • Non-production • No real funds

What Happens After the Decision

From synthetic funding and monitoring to permitted action, portfolio insight and certification.

1 BOOKING & ACTIVATION

Only an authorized, physically funded scenario path enters the account lifecycle.

- **Authorized terms** — Selected amount and governed financing terms.
- **Path identity** — Run, scenario, application, account and advance.
- **Synthetic funding** — Recorded funding—not a real transfer.
- **Source handoff** — Funding dates, keys and retained decision evidence.
- **Account activation** — The exact funded path enters monitoring.

Monitoring begins at the source-owned first expected remittance.

2 DAILY MONITORING LOGIC

On unpaid paths, overlapping signals feed source-defined monitoring status.

OVERLAPPING MONITORING FAMILIES

- **No remittance** — Elapsed days without positive remittance.
- **Sales dormancy** — Consecutive zero-sales days.
- **Liquidity** — Balance, NSF events and negative-balance evidence.
- **Pace** — Cumulative remittance versus capped expectation.
- **Horizon** — Unpaid balance beyond the selected horizon.

NON-CURRENT MONITORING STATUSES

- **WATCH** — Watch-level condition; no higher-priority status.
- **UNDER-PERFORMING** — Adverse pace; no overriding status.
- **SEVERE SHORTFALL** — Severe pace or unpaid horizon overrun.
- **DORMANT NO REMITTANCE** — Source no-remittance threshold reached.

21/30/3 applies to pace; other families have their own triggers.

- **21** — Pace WATCH eligible day 21
- **30** — Pace adverse eligible day 30
- **3** — 3 consecutive eligible-day pace breaches

3 INTERVENTION ROUTES

Recommendations do not grant execution permission.

130 funded scenario paths across two alternatives.

- **No further action** — 126 paid-off paths → **Certified paid-off state**
- **Targeted outreach review** — 3 paths → **Setup + processing permissions** → **Review hold** — Processing held → **Certified held state**
- **Temporary remittance review** — 1 path → **Setup + processing permissions** → **Permitted servicing** — Payments, returns and retries → **Certified account state**

Recommendation ≠ execution. Horizon reached ≠ balance extinguished.

4 WHAT THE SYSTEM PROVES

Monitoring turns daily account behavior into explainable portfolio evidence.

BUSINESS QUESTION	WHAT THE SYSTEM PROVIDES
Is the merchant tracking to plan?	Compare source-owned sales, remittances and expected pace.
How much receivable remains?	Read the balance at the selected observation date and source stage.
Are issues temporary or persistent?	Inspect active families, status and eligible-day persistence.
Which accounts need attention?	Read status, reasons, exposure and permitted servicing state.

Source-owned evidence supports explainable portfolio decisions.

5 PORTFOLIO INTELLIGENCE & CERTIFICATION

Reconciled source evidence feeds certification and analytical consumption.

- **Payment reconciliation** — Match payments, returns, retries and balances.
- **Account-state certification** — Certify permitted account states and retained evidence.
- **Portfolio facts & analysis** — Source-owned exposure, state and scenario comparisons.
- **G3-v2 source certification** — Certify the versioned source-consumption boundary.
- **Power BI P01–P04** — Consume certified facts in analytical views.

Power BI consumes certified facts; it does not create certification.

- **MONITOR** — Read the signals and their context.
- **ROUTE** — Respect review and permission boundaries.
- **EXPLAIN** — Trace evidence to account and portfolio results.
- **CERTIFY** — Reconcile states; retain the evidence.

Signals inform governed action. Reconciled evidence informs portfolio decisions.

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